



REFERENCE

BA5223 № 0000011672

CONTROL NUMBER

P - 0012 - 7301

The purpose of the supplement is to provide sufficient independent data to improve the international «transparency» and fair academic and professional recognition of qualifications (diplomas, degrees, certificates, etc).

It is designed to provide a description of the nature, level, context, content and status of the studies that were pursued and successfully completed by the individual named on the original qualification to which this supplement is appended. It should be free from any value judgements, equivalence statements or suggestions about recognition.

Information in all eight sections should be provided. Where information is not provided, an explanation should give the reason why.

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1. INFORMATION IDENTIFYING THE HOLDER OF THE QUALIFICATION

1.1 Family Name(s)

TREGUBOV

Трегубов

1.2 Given Name(s)

ANDREI

Андрей Сергеевич

1.3 Date of Birth

31 March 2001

1.4 Student Identification Number

BA5223 № 0000011672

2. INFORMATION IDENTIFYING THE QUALIFICATION

2.1 Name of Qualification and Title Conferred

Bachelor

Бакалавр

2.1.1 Date of Award

26 June 2023

26 июня 2023

2.2 Main Field(s) of Study for the Qualification

Mathematics

Математика

2.2.1 Concentration

Mathematics

Математика

2.2.2 Educational Programme

Mathematics

Математика

2.3 Name of Institution Awarding Qualification

National Research University Higher School of Economics (Higher School of Economics/HSE)

Федеральное государственное автономное образовательное учреждение высшего образования «Национальный исследовательский университет «Высшая школа экономики» (Высшая школа экономики/НИУ ВШЭ)

2.4 Name of Institution Administering Studies

Same as in 2.3

2.5 Language(s) of Instruction/Examination

Russian and English

Русский и частично английский

3. INFORMATION ON THE LEVEL OF THE QUALIFICATION

3.1 Level of Qualification

Bachelor

First stage tertiary university level education according to the International Standard Classification of Education (ISCED) (According to the National Qualifications Framework of the Russian Federation - Level 6).

3.2 Official Length of Programme

4 years, 240 credits

4 года, 240 кредитов

3.3 Access Requirements(s)

A Certificate of completed general secondary education and results of the unified state examinations or entrance examinations

4. INFORMATION ON THE CONTENTS AND RESULTS GAINED

4.1 Mode of Study

Full Time Education

4.2 Programme Requirements

The Degree provides a more academically than professionally oriented education. Teaching consists of lectures, seminars, workshops, laboratory works, etc. Forms of assessments: tests, exams, term papers, internships, the Final Assessment (includes the State Interdisciplinary Examination and the public defence of Graduation Thesis). Students are admitted to the Final Assessment after they have achieved passing grades in each subject of the curriculum.

4.3 Programme Details

Содержание программы

Physical Training, Project Proposal, Independent English Language Test and Project Activity are not considered in GPA calculation.

UNIT TITLE / НАИМЕНОВАНИЕ ДИСЦИПЛИНЫ	CREDITS/ КРЕДИТЫ	FINAL GRADE/ ИТОГОВАЯ ОЦЕНКА	ECTS GRADE/ ОЦЕНКА
CURRICULUM (COMPULSORY AND ELECTIVE COURSES) / ПРОГРАММА ОБУЧЕНИЯ (ОБЯЗАТЕЛЬНЫЕ КУРСЫ И ДИСЦИПЛИНЫ ПО ВЫБОРУ)			
Algebra / Алгебра	11	10/отлично	A++
Data Bases and Programming / Базы данных и программирование	5	9/отлично	A+
Safe Living Basics / Безопасность жизнедеятельности	1	8/отлично	A
Introduction to Artificial Intelligence / Введение в искусственный интеллект	3	7/хорошо	B+
Introduction to Nonlinear Dynamics / Введение в нелинейную динамику	4	5/удовлетворительно	C+
Introduction to Topology / Введение в топологию	8	6/хорошо	B-
Geometry / Геометрия	9	6/хорошо	B-
Discrete Mathematics / Дискретная математика	8	10/отлично	A++
Differential Geometry / Дифференциальная геометрия	8	10/отлично	A++
Differential Equations (Studied in English) / Дифференциальные уравнения (изучена на английском языке)	8	10/отлично	A++
Project Proposal (Studied in English) / Защита концепции выпускной квалификационной работы (изучена на английском языке)		7/хорошо	B+
History / История	4	7/хорошо	B+
History of Mathematics / История математики	4	9/отлично	A+
Quantum Mechanics for Mathematicians / Квантовая механика для математиков	6	6/хорошо	B-
Computer Topology / Компьютерная топология	6	10/отлично	A++
Logic and Algorithms / Логика и алгоритмы	4	6/хорошо	B-
Mathematical Physics / Математическая физика	6	9/отлично	A+
Calculus / Математический анализ	24	10/отлично	A++
Mathematical English / Математический английский	3	7/хорошо	B+
Independent English Exam / Независимый экзамен по английскому языку		7/хорошо	B+
The Basics of Web-technologies / Основы web-технологий	5	8/отлично	A
Programming Basics / Основы программирования	5	8/отлично	A
Introduction into Python / Основы программирования на Python	5	9/отлично	A+
Applied theory of dynamical systems (Studied in English) / Прикладная теория динамических систем (изучена на английском языке)	4	7/хорошо	B+
The Development of Web-applications / Разработка web-приложений	5	5/удовлетворительно	C+
Probability Theory / Теория вероятностей	8	10/отлично	A++
Group theory / Теория групп	5	6/хорошо	B-
Field Theory / Теория поля	5	10/отлично	A++
Theory of Complex Functions (Studied in English) / Теория функций комплексного переменного (изучена на английском языке)	7	10/отлично	A++
Number Theory / Теория чисел	3	5/удовлетворительно	C+
Physical Training / Физическая культура	1	зачтено	Passed
Philosophy (Studied in English) / Философия (изучена на английском языке)	4	9/отлично	A+
TERM PAPERS / КУРСОВЫЕ РАБОТЫ			
Statistical Characteristics of the Linear Field of Wind Waves / Статистические характеристики линейного поля ветровых волн	8	8/отлично	A
Rogue Wave Formation on the Surface of Deep Water: Linear Theory / Формирование волны – убийцы на поверхности глубокой воды: линейная теория	8	10/отлично	A++
PROJECT ACTIVITY / ПРОЕКТЫ	5	Passed / зачтено	
INTERNSHIPS, RESEARCH WORKS AND SEMINARS / ПРАКТИКИ И НАУЧНО-ИССЛЕДОВАТЕЛЬСКАЯ РАБОТА			
Research seminar "Material analysis" / Научно-исследовательский семинар "Вещественный анализ"	6	5/удовлетворительно	C+
Research seminar "Optimization Methods" / Научно-исследовательский семинар "Методы оптимизации"	4	8/отлично	A
Research seminar "Basics of the oscillation theory" / Научно-исследовательский семинар "Основы теории колебаний"	7	6/хорошо	B-
Research semina "Operator Theory" / Научно-исследовательский семинар "Теория операторов"	5	6/хорошо	B-
Research seminar "Functional analysis" / Научно-исследовательский семинар "Функциональный анализ"	4	5/удовлетворительно	C+
Graduation Internship / Преддипломная практика	8	10/отлично	A++
GRADUATION THESIS / ВЫПОЛНЕНИЕ И ЗАЩИТА ВЫПУСКНОЙ КВАЛИФИКАЦИОННОЙ РАБОТЫ			
Probability Distributions for Finite Ensembles of Irregular Waves / Вероятностные распределения для конечных ансамблей нерегулярных волн	6	9/отлично	A+
NUMBER OF CREDITS / ИТОГО ПО ОСНОВНОЙ ПРОГРАММЕ	240		
OPTIONAL COURSES / ФАКУЛЬТАТИВНЫЕ ДИСЦИПЛИНЫ			
Academic English Writing / Академическое письмо на английском языке	2	4/удовлетворительно	C-
English Language / Английский язык	10	8/отлично	A
Программирование / Программирование	6	5/удовлетворительно	C+

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4.4 Grading Scheme

ECTS Grades		10-point scale (Exam)	5-point scale	Pass/Fail Test
A++	Excellent	10	отлично (существенно превосходит ожидания)	зачтено (Passed)
A+	Very good	9	отлично (превосходит ожидания)	зачтено (Passed)
A	Very good	8	отлично	зачтено (Passed)
B+	Good	7	хорошо	зачтено (Passed)
B-	Good	6	хорошо	зачтено (Passed)
C+	Satisfactory	5	удовлетворительно	зачтено (Passed)
C-	Satisfactory	4	удовлетворительно	зачтено (Passed)
F	Fail	3	неудовлетворительно	не зачтено (Fail)
F	Fail	2	неудовлетворительно	не зачтено (Fail)
F	Fail	1	неудовлетворительно	не зачтено (Fail)

4.5 Overall Classification

In the Russian Federation Diplomas with Distinction (degrees with honors) are awarded to students who have passed the state examination and defended the final qualifying work perfectly (ECTS - A) and have only good (ECTS - B) and excellent (no less than 75 % of the total) marks for all examinations, which they passed while studying at the university.

5. INFORMATION ON THE FUNCTION OF THE QUALIFICATION

5.1 Access to Further Study

The Degree allows to access the master's degree courses.

5.2 Professional Status

Bachelor's graduates are prepared to be engaged in research and training activities requiring fundamental knowledge and professional skills in mathematics and applied mathematics at enterprises (companies) of all forms of business ownership, education and research organisations.

6. ADDITIONAL INFORMATION

6.1 Additional information

Higher School of Economics was founded on November 20, 1992 in Moscow (Russia) on the initiative of renowned Russian economists and leading reformers. In August 2008 the HSE became one of three Russian Universities operating under the auspices of the Government of the Russian Federation. In October 2009 the Higher School of Economics was awarded the status of a National Research University. Higher School of Economics is licenced to hold educational activities in the field of higher education and accredited by Federal Service for Supervision in Education and Science.

101000, Russia, Moscow,
Myasnitskaya 20,
+7 (495) 621-7983 (tel.),
+7 (495) 628-7931 (fax),
hse@hse.ru

101000, Россия, г. Москва,
ул. Мясницкая, д. 20,
+7 (495) 621-7983 (тел.),
+7 (495) 628-7931 (факс),
hse@hse.ru

6.2 Additional Information Sources

Official website of the National Research University Higher School of Economics (HSE) - www.hse.ru

Official website of the Ministry of Education and Science of The Russian Federation - mon.gov.ru

Official website of the National Accreditation Agency - www.nica.ru

7. CERTIFICATION OF THE SUPPLEMENT

7.1 Capacity

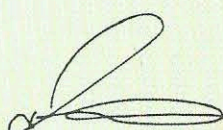
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7.2 Signature

N.Anisimov

7.3 Date of Signature

01 July 2023



Credits. Workload of student's training activity in National Research University Higher School of Economics (HSE) is defined in academic hours and credits. One academic hour is equal to 40 astronomic minutes of students' studies (both in theory and in practice). One credit is equal to 38 academic hours (apr. 25 astronomic hours) depending on complexity of studied discipline.

Academic Performance Rating. There are two forms of final academic performance rating in HSE: examination and test. As a rule educational institutions in Russian Federation apply 5- point grading scale in respect for students' knowledge. Educational outcomes of students in HSE are estimated by 10- point grading scale. Disciplines with final control in test form may also contain the mark in 10-point grading scale. Students of HSE are taught according to annual approved curriculum based on educational standards defining the contents, the forms, the goals and the terms of education. Regulatory time limit for full-time courses is the following: four years for obtaining the bachelor's degree, five years for obtaining the specialist's degree, two years for obtaining the master's degree. Student's pass to the next year of education is possible after getting the "pass" grade or higher in all disciplines of the current year of studies. Students who have completed the plan of studies in full size receive the diploma of higher education and obtain the qualification corresponding to their educational level.

Percentile indicates the share (in a percentage) of students out of an entire selection that demonstrates study results above that of the student in question.

Grade Point Average (GPA) means an indicator showing the average weighted results of all students' grades, whereby the total number of credits for a given course shall serve as the weight coefficient.

8. RUSSIAN EDUCATIONAL SYSTEM

Age	Scientific Degrees	
28	Doctor of Sciences (Level 9)	2-3 years
27	PhD/Candidate of Sciences (Level 8)	
26	Higher Education	
25	Researcher, Lecturer (University education, 180-240 credits, Level 8)	
24	Specialist (Level 7)	Master (Level 7)
23	(University education, 300 - 360 credits)	(University education, 120 credits)
22	5-6 years	2 years
21		Bachelor (Level 6)
20		(University education, 240 credits)
19		4 years
18	Unified State Exam	Secondary Vocational Education
17		Advanced
16	Secondary General Education	Programmes for mid-career professionals
15	2 years	4 years
14		Basic
13		Programmes for skilled workers, junior technicians and employees
12		4 years
11	Basic General Education	5 years
10	Primary General Education	4 years
9		
8		
7		
6		
5		
4	Pre-school General Education	4 years
3		